

# Gitesh Chawda

AI/ML Engineer

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## PROFESSIONAL SUMMARY

Machine Learning Engineer with 3+ years across industry and open source. Author and maintainer of deep learning libraries spanning transformer architectures, Large Language Model (LLM) inference, quantization, and GPU kernel optimization on PyTorch, TensorFlow, and JAX; 48 merged pull requests into the Hugging Face and Keras ecosystems.

## TECHNICAL SKILLS

**Languages and Web:** Python, C++, TypeScript, JavaScript, SQL, React.js, FastAPI, GraphQL, NumPy, Pandas, OpenCV

**ML and Deep Learning:** PyTorch, TensorFlow, Keras 3, JAX, Hugging Face Transformers, Computer Vision, Diffusion Models

**LLM Training and Inference:** vLLM, TensorRT-LLM, Triton, CUDA, XLA, Quantization (INT8/INT4/FP8), LoRA, PEFT, SFT, GRPO, KV Cache Optimization, FlashAttention, Model Serving

**Agentic AI:** LangChain, LangGraph, LlamaIndex, CrewAI, OpenAI Agents SDK, Retrieval-Augmented Generation (RAG), MCP

**MLOps and Cloud:** Docker, Linux, Git, GitHub Actions, CI/CD, AWS (Lambda, API Gateway, S3), GCP, MongoDB, Neo4j, Chroma

## EXPERIENCE

**Creator and Maintainer,** ZeroModels | Open-Source Keras 3 Model Library

Nov 2024 - Present

- Created and solely maintain **ZeroModels**, a backend-agnostic Keras 3 ecosystem of 100+ LLM, VLM, vision, speech, detection, segmentation, and depth architectures across JAX, PyTorch, and TensorFlow.
- Engineered Hugging Face-to-Keras checkpoint conversion with parity validation and native weight-only INT8, INT4, and FP8 quantization built on Keras 3 multi-backend operations.
- Scaled the library to 140,000+ Hugging Face model downloads across 800+ model repositories, with automated testing, releases, documentation, and CI/CD through GitHub Actions.

**Software Developer,** Fiserv | Chennai, India

Jul 2023 - Aug 2024

- Built and maintained a production banking and payments web application on React.js, GraphQL, and AWS serverless (AppSync, Lambda, API Gateway, Cognito, S3).
- Strengthened application security with client-side tampering prevention and server-side request validation, partnering with AppSec teams to remediate vulnerabilities from audits and penetration testing.
- Managed 20+ production deployments and critical hotfixes across development, staging, and production AWS environments.

**Open Source Contributor,** Hugging Face Transformers, Keras, Keras.io, and KerasCV

Apr 2022 - Present

- Enabled batched multimodal audio generation for Qwen2.5-Omni and Qwen3-Omni, fixed Hunyuan-VL preprocessing parity, and decoupled LXMERT and ConvBERT tokenizers from BertTokenizer in Hugging Face Transformers.
- Shipped Keras 3 multi-backend ops (unravel\_index, diagflat, inner, exp2), the PSNR metric API, RandomGrayscale and Equalization layers, the rgb\_to\_grayscale utility, and a RandomHue fix.
- Extended KerasCV with ConvMixer models, CIoU loss, YOLOv8 box-loss support, and RetinaNet backbone tests, plus official keras.io object detection and super-resolution examples.

**Deep Learning Intern,** Braynix AI | Bangalore, India

Sep 2021 - Dec 2021

- Designed deep learning pipelines for medical X-ray abnormality detection, raising validation accuracy 20% over baseline Convolutional Neural Network (CNN) models on 5,000+ images.
- Achieved 92% precision and 0.85+ IoU on medical image segmentation using U-Net with a pretrained EfficientNet-B4 encoder.
- Benchmarked 3 object detection architectures (YOLOv5, SSD, RetinaNet) on 1,000+ labeled medical images at 0.78+ mAP.

## PROJECTS

**ZeroInfer:** Local-first desktop inference platform running 143+ Hugging Face model families (LLM, VLM, diffusion, ASR, TTS, detection, segmentation) fully on-device, with an OpenAI-compatible API with MCP Server.

**Trinix:** Triton-accelerated PyTorch GPU kernel library with fused attention (MHA, GQA, MQA), RMSNorm, SwiGLU, RoPE, and ALiBi, delivering up to 4.8x faster execution.

**OpenRAG:** Self-hosted RAG framework with switchable vector (Chroma) and knowledge graph (Neo4j) retrieval, cross-encoder reranking, and local LLMs.

**txtcode:** TypeScript framework orchestrating AI coding agents across 6 messaging platforms and 10 LLM backends.

## EDUCATION

**Master of Science in Computer Science,** California State University, Fullerton, CA

Aug 2024 - May 2026

**Bachelor of Engineering in Computer Science,** Pimpri Chinchwad College of Engineering | Pune, India

Aug 2019 - Jul 2023

## PUBLICATIONS

**EffiConvRes: An Efficient CNN with Residual Connections and Depthwise Convolutions.** IEEE ICCUBE, 2023.

**Survey of Pre-trained CNN and CNN-Transformer Architectures.** Smart Grid and Electric Vehicle Conference.